

Coupling of polyhydroxybutyrate and zero-valent iron for enhanced treatment of nitrate pollution within the Permeable Reactive Barrier and its downgradient aquifer

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ARTICLE INFO

Keywords:

Denitrification
Aquifer
Groundwater
In-situ remediation
Permeable reactive barrier

ABSTRACT

Permeable Reactive Barriers (PRBs) have been utilized for mitigating nitrate pollution in groundwater systems through the use of solid carbon and iron fillers that release diverse nutrients to enhance denitrification efficiency. We conduct laboratory column tests to evaluate the effectiveness of PRBs in remediating nitrate pollution both within the PRB and in the downgradient aquifer. We use an iron-carbon hydrogel (ICH) as PRB filler, which has different weight ratios of polyhydroxybutyrate (PHB) and microscale zero-valent iron (mZVI). Results reveal that denitrification in the downgradient aquifer accounts for at least 19.5 % to 32.5 % of the total nitrate removal. In the ICH, a higher ratio of PHB to mZVI leads to higher contribution of the downgradient aquifer to nitrate removal, while a lower ratio results in smaller contribution. Microbial community analysis further reveals that heterotrophic and mixotrophic bacteria dominate in the downgradient aquifer of the PRB, and their relative abundance increases with a higher ratio of PHB to mZVI in the ICH. Within the PRB, autotrophic and iron-reducing bacteria are more prevalent, and their abundance increases as the ratio of PHB to mZVI in the ICH decreases. These findings emphasize the downgradient aquifer's substantial role in nitrate removal, particularly driven by dissolved organic carbon provided by PHB. This research holds significant implications for nutrient waste management, including the prevention of secondary pollution, and the development of cost-effective PRBs.

1. Introduction

Agricultural activities, such as crop cultivation and livestock rearing, introduce nitrogenous compounds into the soil. These compounds are then transformed by indigenous soil microorganisms into nitrates, which can contaminate groundwater (Boumaiza et al., 2022; Shu et al., 2022). For example, over 28 % of groundwater resources in China have been found to have nitrate levels exceeding the World Health Organization (WHO)'s standard (i.e., 10 mg/L) (Gu et al., 2013; Zhang et al., 2023). In some agricultural regions in eastern and central China, nitrate concentration have been recorded at around 400 mg/L, nearly 40 times

the WHO standard (Ma et al., 2012). Therefore, it is crucial to reduce nitrate contamination in the groundwater.

The construction of permeable reactive barriers (PRBs) has emerged as an effective in-situ approach for mitigating nitrate contamination in groundwater. PRBs function by supplying specific nutrients to indigenous microorganisms in groundwater, stimulating their activities and enhancing their bioremediation potential (Budania and Dangayach 2023). As nitrate-contaminated groundwater flows through a PRB, its nitrate levels decline. However, because an excess of nutrients is often introduced into the water to achieve effective remediation of nitrate pollution (Jiang et al., 2021; Pang and Wang 2021), a significant amount

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of residual nutrients remains in the water after nitrate levels have reduced sufficiently to comply with groundwater quality standards. Those residual nutrients migrate to the downgradient aquifer of the PRB, possessing the potential to stimulate bioremediation while lacking electron acceptors (nitrate), resulting in nutrient wastage (Yuan et al., 2020). Incorporating the downgradient aquifer of PRB into the design of PRB can reduce the waste of nutrients, while also enhancing the cost-effectiveness of the PRB. After all, the high construction costs are one of the main barriers limiting the widespread application of PRBs. Existing research, however, has primarily focused on denitrification performance within the PRB, with limited considerations on its denitrification potential in downgradient aquifers.

The choice of PRB fillers is one of the key factors determining PRB denitrification performance, with optimal fillers ensuring long-term, stable nitrate pollution remediation. Solid carbon source PHB, a commonly used filler in PRB, can continuously provide organic carbon for indigenous microorganisms, thereby enhancing the bioremediation efficiency of pollutants (Amanat et al., 2021; Baric et al., 2014). Due to strict restrictions on the concentration of organic matter in groundwater, alternative solutions must be sought to maintain the effective operation of denitrification and avoid the occurrence of secondary pollution (Amoako-Nimako et al., 2021; Ma et al., 2023). Zero-valent iron (ZVI) is a potential alternative solid carbon source for enhancing the denitrification activity of denitrifying bacteria. It provides an external electron donor (Fe(II) and H_2) for denitrifying microorganisms through corrosion dissolution, but is limited by issues such as ammonia production (Amoako-Nimako et al., 2021; Zhao et al., 2022).

Coupling ZVI with solid carbon sources, where ZVI substitutes a portion of the solid carbon source as a nutrient provider, can not only reduce the concentration of organic matter released from the filler to the water body but also avoids the reduction of nitrate to ammonia through abiotic reduction due to ZVI, thanks to the biological denitrification process (Naka et al., 2008; Pan et al., 2023; Sun et al., 2021). This strategy can provide an abundance of resources (nutrients) for the microbes, enhancing interactions (cross-feeding, syntrophic, and mutualistic) and functionalization among them. This enables the microbial to maintain a higher activity, thereby achieving higher long-term denitrification performance than when using solid carbon sources alone (Zhou et al., 2022). Furthermore, the reduction of Fe(III) is often enhanced in the presence of organic carbon, which increases the opportunity for microbial for cyclic utilization of Fe(II) (Naka et al., 2006; Strathmann and Stone 2002). This coupling strategy is expected to enhance the remediation of nitrate contamination using PRB. As mentioned earlier, the downgradient aquifer of the PRB is also anticipated to be an important reaction zone. The nutrients released from solid carbon sources and ZVI, such as organic carbon and Fe (II), exhibit different and complex transport behaviors in groundwater, which are expected to have different impacts on the denitrification process within and downgradient aquifer of PRB. Therefore, in this case, the mechanism and performance by which these nutrients enhance the denitrification process within and downgradient of the PRB aquifer needs to be further clarified.

When PRB fillers are directly exposed to groundwater, these fillers react directly with the water, leading to rapid consumption of the fillers and quick release of nutrients. Here we propose an improved structural design that encapsulates the PRB fillers within hydrogels. This offers several advantages. First, this approach prevents direct contact between PRB fillers and groundwater, enabling controlled release of nutrients and extending the lifespan of the fillers (O'Connor et al., 2018). Second, it ensures even dispersion of fillers within hydrogels, mitigating issues of aggregation and consolidation (Bilardi et al., 2023; Meng et al., 2020). Hydrogels, such as polyvinyl alcohol (PVA) (Kaur et al., 2021) and sodium alginate (SA), are non-toxic and cheap (Yu et al., 2018). Upon crosslinking, they form porous structures that demonstrate exceptional thermal stability and chemical resistance, making them ideal scaffolds for fillers.

In the present study, we examine the nitrate remediation performance of coupled PHB-ZVI fillers that are encapsulated within hydrogels, focusing on both the PRB and its downgradient aquifer. We synthesize iron-carbon hydrogel (ICH) using PHB and microscale zero-valent iron (mZVI), and use ICH as PRB fillers. To investigate its performance, we conduct column experiments to monitor the changes in nitrogen concentration along the flow path, thereby examining the denitrification performance of the PRB and its downgradient aquifer. Additionally, we track variations in total dissolved iron (TDI) and dissolved organic carbon (DOC) to monitor the changes in nutrient distribution. Finally, we employ a high-throughput sequencing technology to analyze the microbial community structure within the PRB and its downgradient aquifer. This provides insight into the response mechanisms between nutrients and microorganisms.

2. Materials and methods

2.1. Preparation of ICH

The raw materials for the preparation of the hybrid scaffolds, including PVA, SA, boronic acid, and calcium chloride, were purchased from Sinopharm Chemical Reagent Co., Ltd. The reactive reagents, including PHB and mZVI, were purchased from Shanghai Macklin Biochemical Co., Ltd. and China Metals Metallurgy Research Institute, respectively. All reagents used in this study were of analytical grade and were used without further purification.

To create ICH, 9 g of PVA and 3 g of SA were mixed in distilled water and was diluted to 300 mL. The resulting suspension reacted at 368 K for 2 h. After cooling to room temperature, PHB and mZVI were individually introduced into the mixture according to the feeding ratio as outlined in Table 1. This mixture was then stirred for 1 hour. Subsequently, the mixture was extruded using an injector into a solution comprised of saturated boronic acid and 4 % $CaCl_2$. After immersion at 277 K for 24 h, the resulting particles were thoroughly rinsed with distilled water to eliminate any residual reaction reagents and subsequently dried in a vacuum freeze dryer. The resulting particles, measuring 2–3 mm in size, were denoted as P, PZ, and Z, respectively.

2.2. Column experiments

The column experiments were conducted using cylindrical columns made from polymethylmethacrylate (aka, PMMA), each with a length of 40 cm and an inside diameter of 4 cm. These columns were equipped with four evenly spaced lateral sampling ports as illustrated in Fig. 1. The entire internal volume of the columns was filled with standard sand as the aquifer medium, with a grain size of 0.5 mm. within each column, two distinct zones were established – the PRB and its downgradient aquifer – with a volumetric ratio of 1:1. Among the four sampling ports, two were situated within the PRB, and the remaining two were placed in the downgradient aquifer. The filling procedure was as follows: a filter sintering net was positioned at both ends of the infiltration column, and the ICH was blended with standard sand at a ratio of 3 % of the total mass. This mixture was then layered into the column in 5 cm increments until reaching a depth of 20 cm from the tail. The remaining 20 cm of the column was filled solely with standard sand. Upon completion, the porosity of the infiltration column was approximately 40 %.

Before starting the experiment, microbial inoculation was performed within the infiltration column. In this procedure, indigenous groundwater microbial was employed as the experimental strain. The

Table 1
Feeding ratios for ICH.

	P	PZ	Z
PHB (%)	12	8	4
mZVI (%)	4	8	12

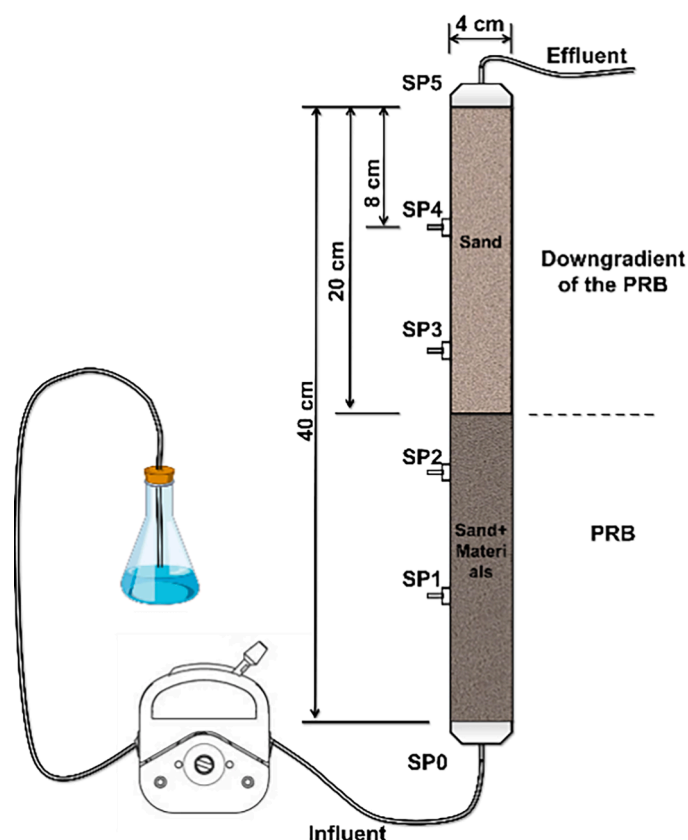


Fig. 1. Schematic diagram of a laboratory column.

inoculating solution was prepared by mixing the indigenous groundwater microbial with simulated groundwater, composed of 0.03 g/L KH_2PO_4 , 0.01 g/L $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, 0.01 g/L $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 0.01 g/L $\text{FeSO}_4 \cdot 2\text{H}_2\text{O}$, 0.36 g/L nitrate, and 1 g/L microelements. The inoculum was delivered into the infiltration column at a rate of 140 $\mu\text{L}/\text{min}$ using a peristaltic pump, and the effluent was recirculated back into the original inoculum for two days, thus completing the inoculation of microorganisms. Following inoculation, a continuous experiment was conducted for 90 days, with simulated groundwater pumped into the column at a rate of 140 $\mu\text{L}/\text{min}$, equivalent to a residence time of 24 h.

2.3. Sample collection

Daily sampling was performed at six designated sampling ports (SP0, SP1, SP2, SP3, SP4, and SP5) along the infiltration columns. The collected samples were filtered through 0.45 μm membrane filters for further analysis. A UV/VIS spectrophotometer (2800 UV/VIS, ONICO, Australia) was used to measure nitrate, nitrite, ammonium, and TDI concentration in solution using the ultraviolet spectrophotometry, N-(1 Naphthalene)-Diaminoethane colorimetric, Nessler's reagent colorimetric, and 1,10-phenanthroline analytical methods, respectively. pH value and dissolved oxygen levels were measured using pH meters (HQ40d Portable pH Meter, Hach Inc., USA). DOC was analyzed by TOC analyzer (Model Sievers InnovOx ES L, Sievers Co., USA). Biological metabolites in the samples were detected using a fluorescence spectrophotometer (FL-4600, Hitachi, Japan). Acetic acid was quantified using a gas chromatography (7890B, Agilent, USA).

2.4. Microbial community analysis using high-throughput sequencing

To assess the microbial community of biofilms, sand samples were collected with sterile spoons from four sampling port sites (SP1, SP2, SP3, and SP4) at the conclusion of the experiment and subsequently

stored at 253 Kelvin until microbial DNA extraction. DNA extraction was carried out using the microbial DNA extraction kit 24 (Biocolors, Shanghai). The V4–V5 regions of 16 S rRNA genes were amplified using 25 primers 338 F (5' - ACTCCTACGGGAGGCAGCAG - 3') and 806 R (5' - GGACTACHVGGGTWTCTAAT - 3'). Illumina MiSeq sequencing was performed at the Shanghai Majorbio Bio-pharm Biotechnology Co., Ltd. (Shanghai, China). Operational taxonomic unit clustering was performed at a 97 % sequence identity threshold. The phylotype coverage, richness, and diversity were estimated by Mothur analysis at the 3 % distance level. Additionally, metagenomic prediction of functional enzymes of the biofilm communities was carried out using the PICRUSt algorithm, utilizing data from the latest KEGG database.

3. Results

3.1. Variation of denitrification performance along the direction of groundwater flow

The denitrification performance along the groundwater flow direction exhibited distinct patterns in the P, PZ, and Z systems, as shown in Fig. 2. In the P system, an nitrate removal efficiency of over 97 % was observed at SP5 during the first 35 days, maintaining a nitrate concentration below 1.3 mg/L (Fig. 2a). During 35–90 days, there was a rapid decline of nitrate removal efficiency, followed by a gradual decrease and stabilization at around 22.3 %, yielding a nitrate concentration of 40.4 mg/L. At 35 days, 76.6 % of the removed nitrate was eliminated between SP0 and SP2, with 6.6 % being removed between SP3 and SP5. This ratio changed to 48.0 % and 32.5 %, respectively, at 90 days. In the PZ system, SP5 exhibited over 97 % of nitrate removal during the initial 20 days (Fig. 2b). Following a rapid then gradual decrease over the next 70 days, nitrate removal efficiency at SP5 stabilized at 28.6 % (37.1 mg/L of nitrate concentration), which was 1.3 times higher than in the P system. On the 90th day, 10.1 mg/L of nitrate was removed between SP0 and SP2 and 4.0 mg/L was removed between SP3 and SP5, accounting for 67.3 % and 26.8 % of the total removed nitrate, respectively. In the Z system, the initial nitrate removal efficiency at SP5 was significantly lower than in the P and PZ systems, at 25.9 %, during the first 5 days (Fig. 2c). Over the next 85 days, nitrate removal efficiency at SP5 slowly increased to 39.7 %, resulting in a nitrate concentration of 31.5 mg/L, 1.77 times that of the P system. On the 90th day, 15.1 mg/L of nitrate was removed in SP1 and SP2 and 3.9 mg/L was removed in SP3 and SP4, accounting for 73.6 % and 19.5 % of the total nitrate removed, respectively.

The nitrate removal process is accompanied by the accumulation of nitrite and ammonium (Fig. S1). In the P system, the nitrite concentration at SP5 consistently remained below 0.2 mg/L, and the ammonium concentration were also relatively low, below 0.7 mg/L (Fig. S1a and d). Observing from SP4 to SP1 along the flow direction, almost all these byproducts were generated in the PRB. In the PZ system, nitrite concentration at SP5 remained low for the first 20 days, then gradually increased to 1.1 mg/L (Fig. S1b). Ammonium concentration at SP5 mirrored the initial low levels of the P system for the first 20 days (Fig. S1e). Subsequently, ammonium concentration abruptly increased to around 2.0 mg/L and eventually stabilized at about 1.5 mg/L, accounting for at least 9.4 % of the reduction product. Observing from SP4 to SP1 along the flow direction, it was found that nitrite and ammonium were predominantly generated within the PRB. In the Z system, nitrite concentration at SP5 consistently remained at around 1.1 mg/L (Fig. S1c). Ammonium concentrations initially remained low for the first 5 days, then sharply increased to around 2.4 mg/L before stabilizing at about 1.5 mg/L, representing over 6.3 % of the reduction product (Fig. S1f). As with the other systems, nitrite and ammonium predominantly originated from the PRB.

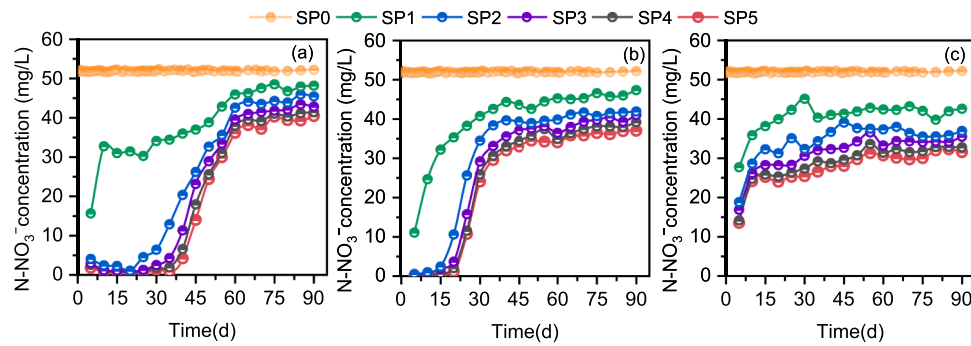


Fig. 2. N-NO_3^- concentration and removal efficiency along the direction of groundwater flow: (a) P system, (c) PZ system, (c) Z system.

3.2. Variation of nutrients along the direction of groundwater flow

The concentrations of DOC and TDI, representing organic and inorganic nutrients, respectively, offer insights into nutrient variations within the P, PZ, and Z systems (Ma et al., 2023; Sun et al., 2021). Over 0–90 days of the operational period, the P system exhibited significant variations in DOC concentration at SP5. Initially, DOC levels rapidly increased to 148.3 mg/L, followed by a gradual decrease and stabilization at approximately 23.4 mg/L (Fig. 3a). Within the first 35 days, DOC concentrations at SP5 to SP0 decrease monotonically. At 35 days, DOC concentrations at SP4 to SP1 were 1.03, 0.94, 0.67, 0.42 times that of SP5, respectively. This reveals that the main variation of DOC concentration occurred within the PRB, while DOC concentrations in the downgradient aquifer exhibit less noticeable changes. In the subsequent 55 days, DOC concentrations showed a trend of initial increase and then decrease with the groundwater flow direction (SP0 to SP5). At 90 days, DOC concentrations at SP4 to SP1 were 1.08, 1.41, 1.32, and 0.40 times that of SP5, respectively (Fig. 3d). In the PZ system, the trend in DOC concentration mirrored that of the P system, with an initial rapidly increase to 131.8 mg/L within the first 15 days, followed by gradual decrease and stabilization around 18.3 mg/L at 90 days (Fig. 3b). DOC concentrations at SP4 to SP1 were 1.02, 0.95, 0.83, and 0.55 times that of SP5, respectively. At 90 days, variations in DOC concentration within the downgradient aquifer of the PRB were less pronounced compared to the P system, with DOC concentrations at SP4 to SP1 being 1.06, 1.24, 1.25, and 0.58 times that of SP5, respectively (Fig. 3e). In the Z system,

DOC concentrations were consistently the lowest among the three systems during the operational period, decreasing from a maximum of 79.5 mg/L to around 13.3 mg/L at SP5 (Fig. 3c). At 90 days, changes in DOC concentration within the downgradient aquifer of the PRB were less prominent compared to the P and PZ systems, with DOC concentrations at SP4 to SP1 were 1.09, 1.17, 0.97, and 0.43 times that of SP5, respectively (Fig. 3f).

Now we turn to TDI. Within the P system, TDI concentrations at SP5 exhibited an upward trend for the first 30 days, peaking at 5.5 mg/L (Fig. 4a). Subsequently, a sharp decline was observed, with concentration stabilizing at approximately 0.69 mg/L between days 40 and 90. This concentration was lower than the initial TDI at SP0, which was measured to be 1.67 mg/L. Before the sharp decrease, the concentrations followed the order $\text{SP3} > \text{SP4} > \text{SP5} > \text{SP2} > \text{SP1}$. Interestingly, the differences between the concentrations at SP3, SP4, and SP5 were within 10 %, indicating that the primary variations in TDI occurred within the PRB. Following the sharp decrease, the concentrations at SP5 to SP1 were consistently lower than the initial concentration (SP0), with the sequence $\text{SP3} > \text{SP2} > \text{SP1} > \text{SP4} > \text{SP5}$. At this stage, the concentration at SP4 was 1.22, 1.51, 1.39, and 1.35 times that of SP5 (Fig. 4d). In the PZ system, the maximum TDI was observed (Fig. 4b). The concentration at SP5 rapidly increased from 0 to 10 days, reaching 23.1 mg/L, then sharply declined to 1.38 mg/L at 35 days. Subsequently, the concentration stabilized at approximately 0.57 mg/L between 40 and 90 days. During this phase, the concentrations at SP4 to SP1 were 1.28, 2.15, 1.88, and 1.71 times that of SP5, respectively (Fig. 4e). In contrast, the

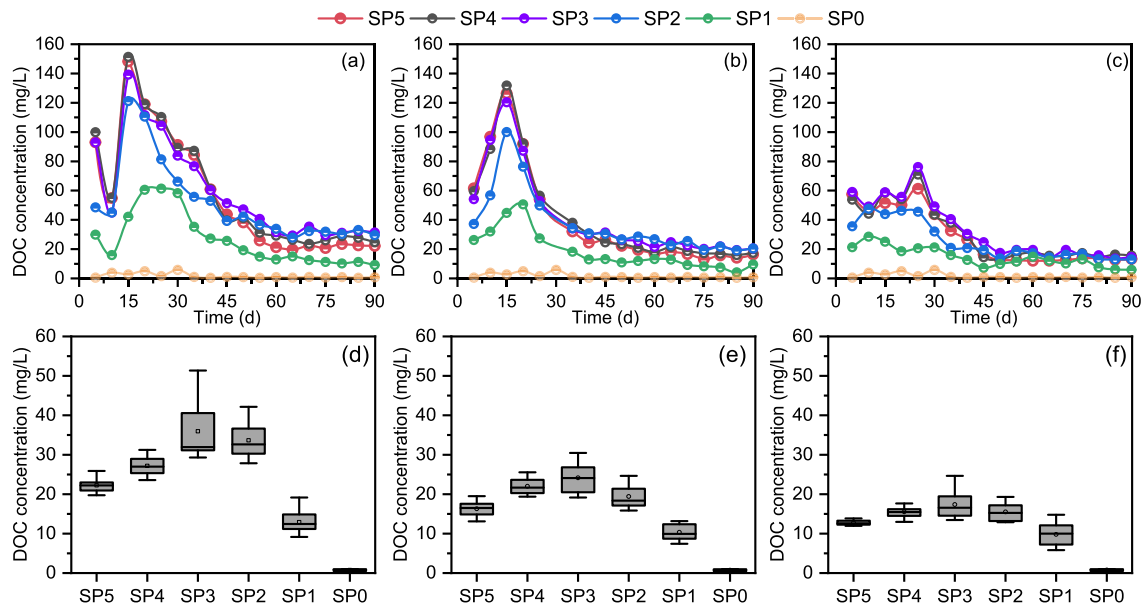


Fig. 3. DOC concentration along the direction of groundwater flow: (a) P system; (b) PZ system; (c) Z system, and 45–90 d (d) P system; (e) PZ system; (f) Z system.

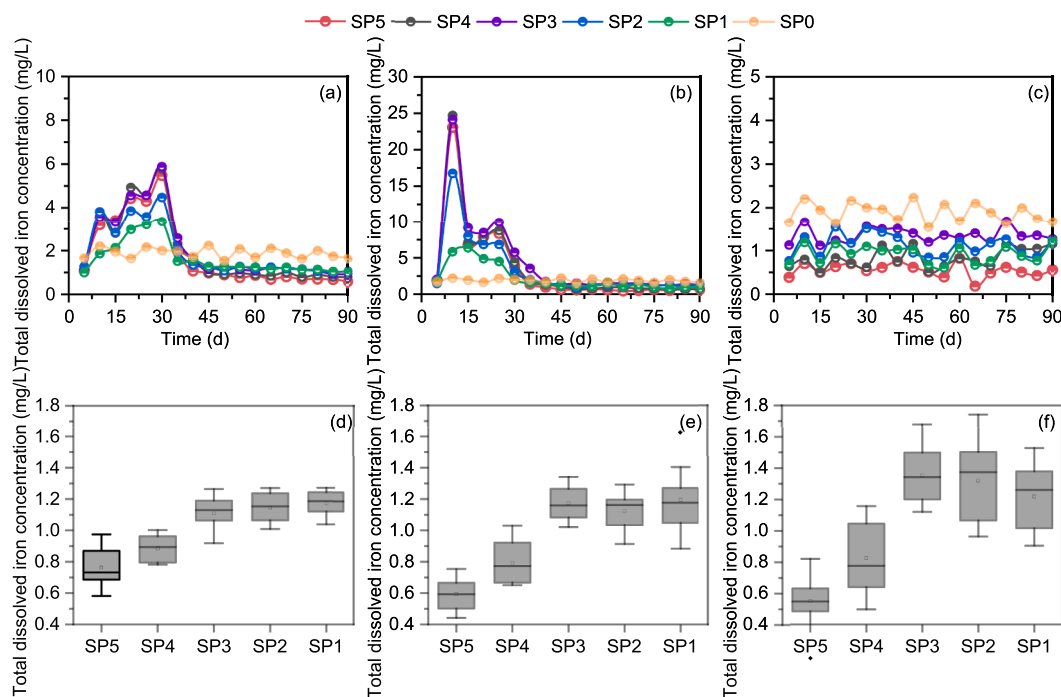


Fig. 4. TDI concentration along the direction of groundwater flow 0–90 d: (a) P system; (b) PZ system; (c) Z system, and 45–90 d (d) P system; (e) PZ system; (f) Z system.

TDI in the Z system remained consistently lower than the initial concentration (SP0) (Fig. 4c). At SP5, the concentration stabilized at 0.55 mg/L, while the concentrations at SP4 to SP1 were 1.49, 2.45, 2.02, and 1.69 times that of SP5, respectively (Fig. 4f).

3.3. 3D-EEM fluorescence spectra under different operational time

We deployed 3D-EEM spectra (Fig. 5) to analyze the dissolved organic matter (DOM) derived from bacterial cultures, with specific peaks denoted as a (Ex=250–280 nm, Em<380 nm) and b (Ex=350 nm, Em=430 nm), representing tyrosine and humic acid, respectively (Xie et al., 2022; Zhang et al., 2022). Importantly, the fluorescence intensity of tyrosine has been found to be closely associated with microbial activity, where higher fluorescence intensity indicates increased microbial activity (Liu et al., 2021). At 15 days, peak a exhibited significantly higher fluorescence intensity in both the P and PZ systems compared to the Z system, with the P system showing slightly higher intensity than the PZ system. This suggests significantly elevated microbial activity in the P and PZ systems in contrast to the Z system. At 40 days, peak a's fluorescence intensity notably decreased in both the P and PZ systems, while it increased in the Z system. Consequently, the microbial activity ranked in the order of P system > Z system > PZ system. Finally, at 90 days, the fluorescence intensity of peak a decreased once more in all three systems. At this time, the Z system exhibited the highest fluorescence intensity for peak a, with the P and PZ systems displaying comparatively lower intensities. Additionally, peak b displayed a high fluorescence intensity exclusively in the Z system, representing the apoptotic products of microorganisms and confirming the increased toxicity of mZVI in the Z system (Marhuenda-Egea et al., 2007).

3.4. Microorganisms community diversity analysis

We employed high-throughput 16S rRNA gene sequencing to explore the microbial community diversity in the P, PZ, and Z systems, with four sampling sites considered (Fig. S2). To facilitate a comparison of microbial communities within the PRB and its downgradient aquifer, SP3 and SP4 were categorized as PRB, and SP1 and SP2 as Downgradient for

analysis. The dominant phylum across all samples was *Proteobacteria* (74.7–94.5 %), followed by *Bacteroidota* (1.31–23.7 %) and *Firmicutes* (0.05–21.2 %) (Fig. 6). *Proteobacteria* was known for its association with denitrification (Huang et al., 2022). *Bacteroidota*, a recognized chemolithotrophic denitrifying bacterium (Miao et al., 2022), was predominantly detected in the P system, constituting 23.7 % and 5.98 % in P-Downgradient and P-PRB, respectively. Conversely, it only accounted for 4.56 %, 1.93 %, 3.52 % and 1.31 % in PZ-Downgradient, PZ-PRB, Z-Downgradient, and Z-PRB, respectively. Interestingly, the relative abundance of *Bacteroidota* was higher in the Downgradient than in the PRB across all samples. *Firmicutes* showed opposing trends, primarily detected in the Z system, with proportions of 1.82 % and 21.2 % in Z-Downgradient and Z-PRB, respectively. In contrast, it accounted for 2.08 %, 7.75 %, 0.68 % and 4.48 % in PZ-Downgradient, PZ-PRB, P-Downgradient and P-PRB, respectively. This phylum is often the predominant species in denitrifying systems with zero-valent iron and plays a role in the reduction of Fe (III) and nitrates (Wang et al., 2009).

The major genus in all samples included *Brevundimonas*, *Dechlorosoma*, *Thermomonas*, *Dechloromonas*, *Simplicispira*, *Azospira*, and *Clostridium sensu stricto 7* (Fig. 6b). The relative abundance of *Thermomonas* was highest in the Z system, followed by the PZ system, and lowest in the P system, and was more concentrated in Downgradient of all samples. *Thermomonas* has been reported to facilitate the coupling of Fe (II) oxidation and denitrification in mixed nutrient denitrification systems and exhibits an autotrophic metabolic pathway, showing a negative correlation with COD values (Han et al., 2023; Pan et al., 2023). *Dechlorosoma*, *Dechloromonas*, *Simplicispira*, and *Azospira*, frequently detected in mixed-nutrient denitrification systems, are known for their key roles in heterotrophic denitrification and for mediating Fe(II) for denitrification (Cheng et al., 2022; Liang et al., 2022; Peng et al., 2021). Their relative abundance was higher in the P (55.1 and 64.8 %) and PZ (42.7 and 37.1 %) systems compared to the Z system (10.4 and 9.9 %). *Flavobacterium*, a common heterotrophic denitrifying bacterium (Li et al., 2022), was predominantly observed in the P system, with relative abundances of 9.7 % and 20.8 % in the PRB and Downgradient, respectively. This genus did not dominate the microbial community in other samples, with relative abundance below 2 %. *Brevundimonas*,

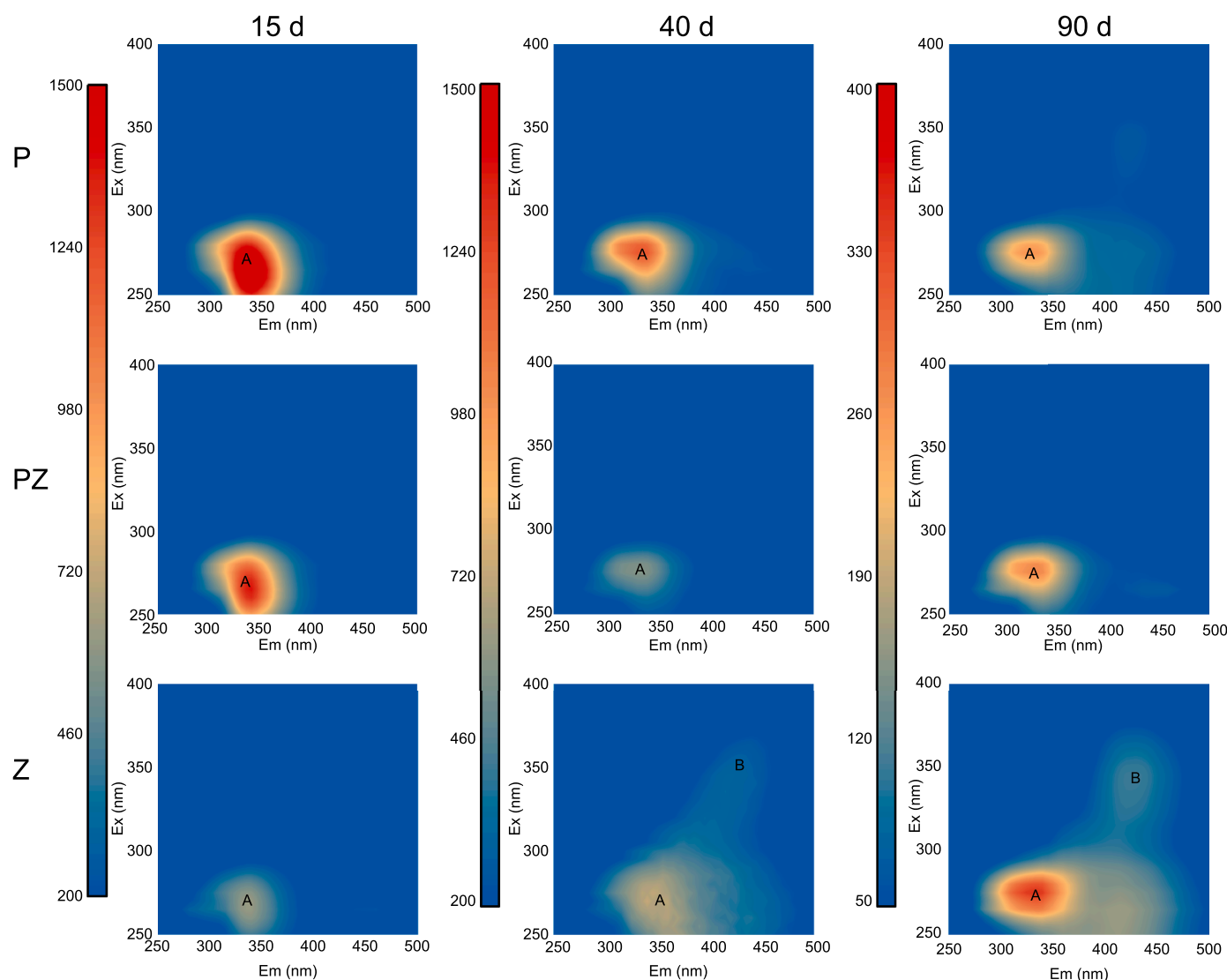


Fig. 5. EEM spectra of SP5 from the P, PZ, and Z systems at 15, 40, and 90 days.

associated with survival under oligotrophic conditions and participation in complex substrate biotransformation, nitrate reduction, and Fe (III) reduction in anaerobic environments (Hu et al., 2023; Wu et al., 2022), was mainly concentrated in the Z and PZ systems (30.6 % and 18.9 %, respectively). Similarly, *Clostridium sensu stricto* 7, belonging to the *Firmicutes* phylum, displayed comparable functionalities and was frequently found in surface water (Zhou et al., 2022). It mainly enriched in the Downgradient, with relative abundances of 2.47 %, 12.5 %, and 33.6 % in Downgradient of the P, PZ, and Z systems, respectively.

4. Discussion

4.1. Effect of DOC and TDI on nitrate removal

In all three PRB systems, the degradation of PHB was the primary source of dissolved organic carbon (DOC), and the main components of DOC were short-chain fatty acids such as acetic acid (Fig. S3), consistent with previous research findings (Amanat et al., 2021). The degradation of PHB initially occurs in its amorphous phase, followed by the crystalline phase. Compared to the crystalline phase, the amorphous phase degrades faster and releases a higher concentration of DOC (Abe et al., 1995; Iggu et al., 2015). Additionally, according to the co-substrate theory, microorganisms prefer to use DOC for denitrification over

other electron donors. It was found that high concentrations of both DOC and total Fe were present in the biofilter at 30 days, but the denitrification pathway was heterotrophic denitrification (Zhou et al., 2022). Based on these findings, we propose that during the initial operation of the three systems, the rapid degradation of the amorphous phase leads to high DOC concentrations, which significantly enhances heterotrophic denitrification. The DOC concentration above 93.2 mg/L resulted in over 97 % nitrate removal. However, when DOC concentration falls below this threshold, nitrate removal efficiency sharply declines. The Z system consistently maintained a DOC concentration lower than 93.2 mg/L, even during the initial operation, resulting in significantly lower nitrate removal efficiency compared to the P and PZ systems. As the amount in the amorphous phase decreased, there was a sharp drop in DOC concentration, accompanied by a weakened microbial activity, leading to a decline in nitrate removal efficiency (Fig. 5).

The majority of nitrate removal occurred within the PRB, primarily due to the high nutrient concentration in this region. Additionally, the presence of ICH as a carrier within the PRB facilitates biofilm formation, promoting the propagation of microorganisms (Russel and Bhaskaran 2023). It is anticipated that DOC would transport to downgradient aquifer of the PRB and stimulate denitrification activity in this region's indigenous microorganism, leading to a positive correlation between DOC concentration and denitrification efficiencies in the downgradient

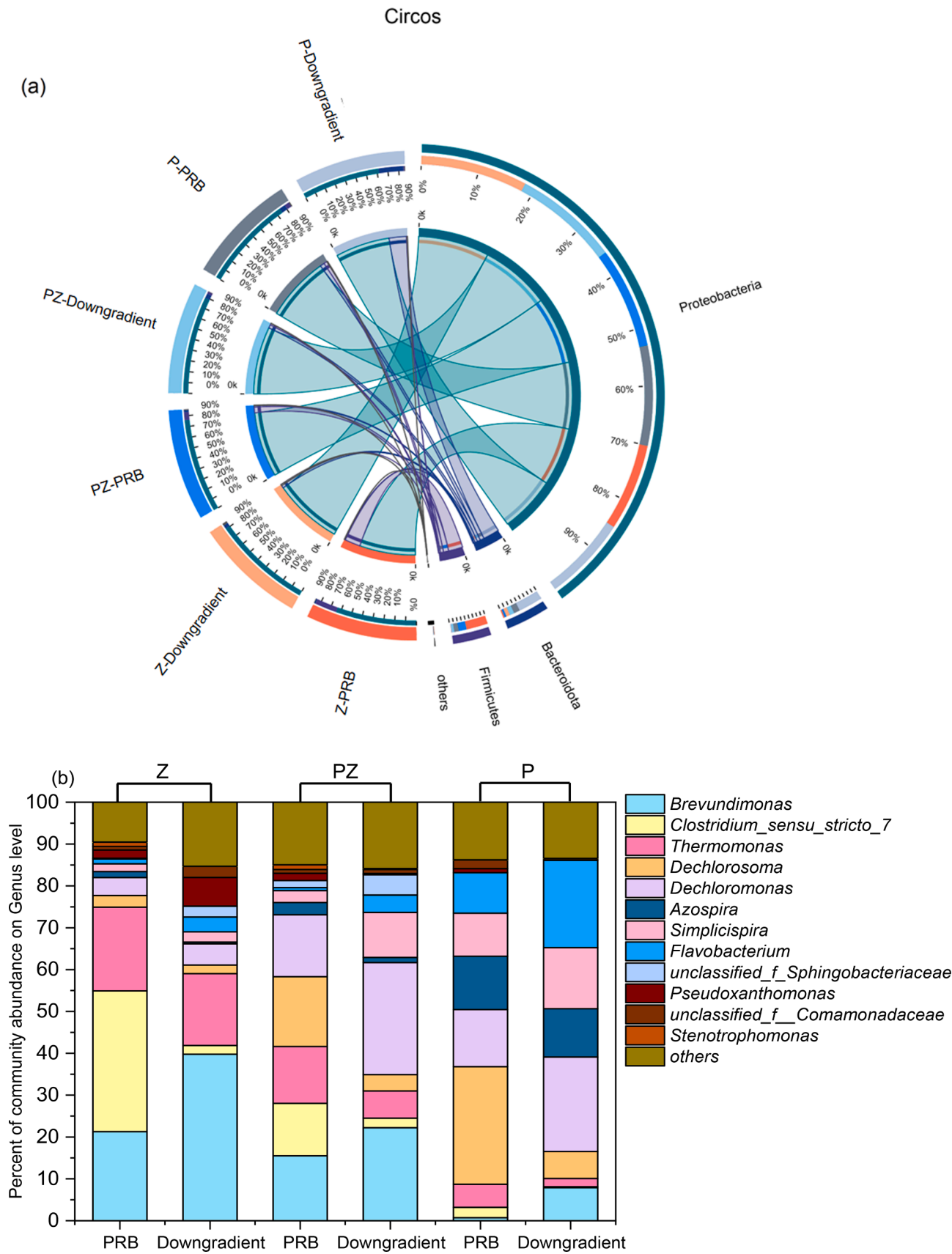


Fig. 6. (a) Structure of the main phylum; (b) genus level of microbial communities of the P, PZ, and Z systems within the PRB and its downgradient aquifer.

aquifer of the PRB. When PHB degradation occurred in the amorphous phase, corresponding to high DOC concentrations, only up to 10 % of removed nitrates were eliminated in downgradient aquifer. Conversely, when PHB degradation occurred in the crystalline phase, corresponding to lower DOC concentrations, approximately 19.5–32.5 % of removed nitrates were eliminated in the downgradient aquifer. This is because, at high DOC concentrations, only a limited residual concentration of nitrates could transport to downgradient aquifer of the PRB alongside DOC, especially in the P and PZ systems. In contrast, a higher residual concentration of nitrates could transport to the downgradient aquifer alongside DOC and undergo denitrification. In all three systems, higher PHB content was associated with a more significant decrease in DOC concentration in the downgradient aquifer, resulting in higher nitrate removal efficiency in this region. This indicates that DOC is a key factor influencing nitrate removal efficiency in the downgradient aquifer of the PRB.

In the early stages, DOC concentration dominated the nitrate removal process in all three systems, even in the presence of high concentrations of total dissolved iron (TDI) detected in the P and PZ systems. This phenomenon aligns with the “co-substrate cultivation” theory, suggesting that microorganisms tend to utilize organic carbon for energy production over other substrate (Wang et al., 2023). In addition, heterotrophic denitrification exhibits higher denitrification efficiencies than denitrification using TDI, because heterotrophic denitrifying bacteria have a higher specific growth rate than bacteria supported by TDI (Pang and Wang 2021). Despite the lower ZVI content in the P and PZ systems compared to the Z system, they contain much higher concentrations of TDI. This can be explained by the higher ZVI content in ICH, resulting in greater cell toxicity, which inhibits microbial activity and restricts biological corrosion of iron (Peng et al., 2021; Zhu et al., 2023).

In the P and PZ systems, high microbial activity enabled biological corrosion to release high concentrations of TDI. However, as microbial activity decreased over time, biological corrosion significantly reduced, and TDI mainly resulted from the reaction of iron with nitrate, dissolved oxygen (DO), and water (Kim and Cha 2021; Strathmann and Stone 2003). Currently, TDI concentration was positively correlated with the ZVI content in the ICH. As DOC concentration decreased, support for heterotrophic denitrification diminished, and TDI concentration became more important for nitrate removal. This suggests a shift from heterotrophic denitrification to iron-related denitrification in response to environmental changes (Zhou et al., 2022). The shift in the dominance of factors controlling the fluorescence intensity of tyrosine in all the three systems—from DOC to TDI—also validates the transition in microbial communities. Ma et al. designed an iron/activated carbon hydrogel system that effectively promotes iron-supported denitrification under low carbon-to-nitrogen ratios and limits ammonium generation (Ma et al., 2023). This system behaved similarly to the P, PZ, and Z systems, indirectly confirming the potential of ICH to enhance iron-supported denitrification.

At low DOC concentrations, the ZVI content in ICH enhanced nitrate removal primarily within the PRB, with limited impact on the downgradient aquifer. This phenomenon could be explained as follows. First, under anaerobic conditions, the primary form of TDI is Fe^{2+} , which, following microbial oxidation, transforms into the more easily precipitable Fe^{3+} , leading to precipitation within the PRB. Second, the production of TDI is finite, being rapidly oxidized and precipitated before transport. In all three systems, TDI concentrations within the PRB were lower than the initial concentrations (SP0), confirming inadequate supply of TDI. The presence of orange-red precipitates observed within the PRB further confirmed iron oxidation. The limited concentration of residual TDI transported to the downgradient aquifer is utilized by indigenous microorganisms for denitrification, as evidenced by the observed decrease in TDI concentrations in the downgradient aquifer. The higher the ZVI content in the ICH, the more significant the decrease in TDI is observed in the downgradient aquifer. Although the

contribution of PRB downgradient aquifer in the Z system to nitrate removal is lower than that of the P and PZ systems, the removed nitrate concentrations are still higher than those in the P and PZ systems. This suggests that TDI plays a significant role in nitrate removal at low DOC concentrations.

4.2. Responses of microorganisms to DOC and TDI, and key mechanisms of nitrate removal

Compared to indigenous microorganisms (Fig. S4), the introduction of ICH selectively altered the indigenous microbial community structure. Although the microbial populations in the three systems were remarkably similar, they exhibited variations based on DOC and TDI levels. Most microorganisms were facultative, including *Dechlorosoma*, *Dechloromonas*, and *Simplicispira*. The dominance of facultative microorganisms over heterotrophic ones suggests a shift in the nutritional structure of microorganisms from heterotrophic to facultative as DOC concentrations decrease. Furthermore, the fluorescence intensity of tyrosine in the Z system at 45 days was higher than that in the PZ system, suggesting that TDI may accelerate this shift.

Due to the organic substrate preference of facultative microorganisms, their relative abundance was positively correlated with DOC concentration in the ICH system. Heterotrophic microorganisms like *Flavobacterium* followed the same trend. As the ICH within the PRB serves as the primary source of DOC, one might expect both facultative and heterotrophic microorganisms to be predominantly concentrated within the PRB itself. However, in all the three systems, heterotrophic microorganisms were mainly enriched in the downgradient aquifer, while facultative microorganisms showed higher relative abundance both within the PRB and in its downgradient aquifer. Considering the observed nitrate removal efficiency, these findings suggest that DOC is favorable in promoting biologically mediated denitrification in the downgradient aquifer of the PRB under long-term operation. Additionally, the autotrophic microorganisms including *Thermomonas*, and iron-reducing bacteria such as *Brevundimonas* and *Clostridium sensu stricto* 7, mutually depend on each other, forming an iron cycling system. Their relative abundance exhibited a positive correlation with the ZVI content in the ICH, and was primarily concentrated in the downgradient aquifer of the PRB. The abundance of key genes involved in the Fe(II)/Fe(III) oxidation–reduction cycle, based on KEGG functional gene profiling, also showed similar results (Fig. S5). These observations reveal microbes’ response to the limited transport of TDI, with iron-related microorganisms displaying a propensity to inhabit within the PRB.

In summary, the nitrate removal pathway within the PRB system is primarily facilitated by biological denitrification, while the contribution from non-biological reduction offered by iron is limited (less than 10 %). In the initial stages of operation, heterotrophic denitrification is the dominant removal mechanism, effectively reducing nitrate levels. However, as the operation progresses, the diminishing DOC concentration leads to a reduction in the efficiency of heterotrophic denitrification. At this point, the nitrate removal rate is primarily dictated by autotrophic denitrification mediated by ferrous ions, with the potential for iron cycling serving as the key determinant of nitrate removal efficiency.

4.3. Potential of solid carbon source and ZVI in PRB remediation

It is evident that the release of DOC and TDI from solid carbon and ZVI can transport to downgradient aquifer of the PRB and enhance the denitrification activity. In comparison to TDI, DOC exhibits greater transport capacity over longer distances; however, its denitrification efficiency is relatively modest. Therefore, it is better suited for addressing nitrate pollution characterized by larger-scale, lower-concentration contamination, and the breadth of the PRB can be adjusted suitably. Nitrate removal processes involving the iron cycle yield better results, due to higher microbial activity. However, the limited transport

behavior of TDI, with the vast majority of nitrate removal occurring within the PRB, makes it suitable for addressing localized, higher-concentration nitrate contamination. A higher proportion of ZVI in the ICH leads to improved denitrification efficiency. However, when the content of solid carbon sources is much lower, the benefits of enhanced biological denitrification effects might not be apparent. This is primarily owes to the fact that, during the initial stages of operation, the metabolic growth of denitrifying microbes supported by ZVI is sluggish. Only when an optimal balance of solid carbon sources is maintained can an effective, long-term denitrification outcome be realized. Fig. S6 illustrates the nitrate removal performance and variations in DOC/TDI levels when materials devoid of solid carbon sources are employed as PRB fillers. In this scenario, nearly all nitrate is removed through abiotic reduction by ZVI, resulting in poor biological denitrification performance.

5. Conclusion

In the present study, we used ICH as the PRB fillers and conducted one-dimensional column experiments to investigate denitrification performance and mechanisms within the PRB and in its downgradient aquifer. We summarize the key findings as follow:

1. The DOC and TDI released from ICH can transport to the down-gradient aquifer of the PRB, providing downgradient aquifer denitrification that accounts for at least 19.5 % to 32.5 % of the total removed nitrate under the conditions examined in the present study.
2. The PRB serves as the sole supply zone for DOC and TDI. The DOC concentrations significantly increased within the PRB and gradually decreased in the downgradient aquifer. In contrast, TDI did not exhibit an upward trend within the PRB and showed a sudden concentration decrease in the downgradient aquifer. This reveals the limited transport capacities of TDI compared to DOC, resulting in a lower contribution of TDI to nitrate removal in the downgradient aquifer of the PRB.
3. The transport behaviors of DOC and TDI caused corresponding responses in microbial communities. Iron cycling microorganisms associated with TDI were predominantly concentrated within the PRB. Conversely, heterotrophic and mixotrophic microorganisms related to DOC were concentrated in the downgradient aquifer of the PRB.

In summary, solid carbon source is better suited for addressing nitrate pollution characterized by larger-scale, lower-concentration contamination, and the breadth of the PRB can be adjusted suitably. Conversely, ZVI demonstrates limited potential for denitrification in the downgradient aquifer of the PRB and is more suitable for mitigating localized, higher-concentration nitrate contamination. Our investigations provide a theoretical foundation for the potential design of a more cost-effective PRB system that combines solid carbon sources and ZVI to remediate nitrate-contaminated groundwater.

CRediT authorship contribution statement

Wenhao Yu: Writing – original draft, Writing – review & editing, Formal analysis. **Tianyuan Zheng:** Conceptualization, Methodology, Funding acquisition, Resources, Writing – review & editing, Supervision. **Bo Guo:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision. **Yiheng Tao:** Writing – review & editing. **Lecheng Liu:** Writing – review & editing, Supervision. **Ni Yan:** Writing – review & editing, Supervision. **Xilai Zheng:** Supervision, Project administration, Funding acquisition, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

All of the data used in this study are reported in the paper.

Acknowledgments

This work was supported by the key project of Joint Foundation of NSFC-Shandong (No. U1806210) and National Natural Science Foundation of China (No. 42272282).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2023.121060](https://doi.org/10.1016/j.watres.2023.121060).

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